

Alexandra Pevzner

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Employment

Zelevinsky Postdoctoral Fellow, Northeastern University 2024-2027

Education

University of Minnesota, Twin Cities.....

Ph.D. in Mathematics

Advisor: Professor Victor Reiner 2019-2024

Thesis: *Symmetric quotients of polynomial rings and Stanley–Reisner rings*

M.S. in Mathematics

Masters/oral exam topic: *Invariant theory of finite groups* 2019-2021

B.S. in Mathematics

with High Distinction 2013-2017

Research Interests

Commutative algebra and combinatorics, with an emphasis on invariant theory and free resolutions.

Publications

Alexander duals of symmetric simplicial complexes and Stanley–Reisner ideals ☞

with Ayah Almousa, Kaitlin Bruegge, Martina Juhnke-Kubitzke, and Uwe Nagel

Advances in Applied Mathematics, Volume 178 (2026)

Symmetric group fixed quotients of polynomial rings ☞

Journal of Pure and Applied Algebra, Volume 228, Issue 4 (2024)

Equivariant resolutions over Veronese rings ☞

with Ayah Almousa, Michael Perlman, Victor Reiner, and Keller VandeBogert

Journal of the London Mathematical Society, Volume 109, Issue 1 (2024)

Preprints.....

Syzygies of the transfer ideal of the symmetric group ☞

with Harm Derksen

Teaching

Instructor of Record.....

Northeastern University

Math 5112: Algebra 2 (graduate level) S26

Math 1365: Introduction to Mathematical Reasoning F25

University of Minnesota

Math 1271: Calculus I S24, F18

Math 2263: Multivariable Calculus S22

Teaching Assistant.....

University of Minnesota

Math 1271: Calculus I S23, F19

Math 2263: Multivariable Calculus	F21, S20
Math 3283W: Sequences, Series, and Foundations	F23, F22, S21, F20
Calculus III for Talented Youth Mathematics Program (UMTYMP)	F22

University of Minnesota REU in Combinatorics and Algebra (Teaching Assistant).....

Virtual resolutions of points in $\mathbb{P}^n \times \mathbb{P}^m$ 2023

Mentor: Prof. Christine Berkesch

Students: Isidora Bailly-Hall, Karina Dovgodko, Sean Guan, Sai Sivakumar, Jishi Sun

Invariant theory of cyclic groups 2021

Mentor: Prof. Victor Reiner

Students: Swapnil Garg, Frank Lu, Kevin Ren, Brian Sun

Topology of augmented Bergman complexes 2021

Mentor: Prof. Victor Reiner

Students: Elisabeth Bullock, Aidan Kelley, Kevin Ren, Gahl Shemy, Dawei Shen, Brian Sun, Amy Tao, Joy Zhang

Peer-Assisted Learning at University of Minnesota.....

Physics 1301W: Mechanics F16

Physics 1302W: Electricity and Magnetism S16, S17

Research Talks

Special Session on Topics in Commutative Algebra

AMS Spring Central Sectional Meeting, North Dakota State University April 2026

Commutative Algebra Seminar

Purdue University April 2026

Special Session on Homological and Combinatorial Commutative Algebra

AMS Spring Eastern Sectional Meeting, Boston College March 2026

Syzygies and Mirror Symmetry Seminar

Virtual seminar March 2026

Special Session on Syzygies and Local Cohomology in Commutative Algebra

AMS Fall Southeastern Sectional Meeting, Tulane University October 2025

Special Session on Algebraic Combinatorics and Combinatorial Commutative Algebra

AMS Fall Southeastern Sectional Meeting, Tulane University October 2025

Special Session on Algebraic and Enumerative Combinatorics

CMS Summer Meeting, Université Laval June 2025

Special Session on Homological Methods in Commutative Algebra

AWM Research Symposium, UW Madison May 2025

Special Session on Commutative Algebra and Interactions with Combinatorics

AMS Spring Eastern Sectional, Hartford April 2025

Combinatorics Seminar

Brandeis University Nov 2024

Geometry, Algebra, Singularities, and Combinatorics Seminar

Northeastern University Oct 2024

Geometry, Physics, and Representation Theory Seminar

Northeastern University Oct 2024

Combinatorics Seminar

Brown University Sept 2024

Special Session on Group Actions in Commutative Algebra

JMM San Francisco Jan 2024

Gender Equity in the Mathematical Study (GEMS) of Commutative Algebra workshop <i>University of Minnesota, Twin Cities</i>	Nov 2023
Algebra Seminar <i>University of Nebraska, Lincoln</i>	Nov 2023
Special Session on Commutative Algebra, Differential Operators, and Singularities <i>AMS Fall Central Sectional, Creighton University</i>	Oct 2023
Special Session on Combinatorial and Homological Methods in Commutative Algebra <i>AWM Research Symposium, Clark Atlanta University</i>	Sept 2023
Upcoming Researchers in Commutative Algebra (URiCA) conference <i>University of Nebraska, Lincoln</i>	May 2023
Combinatorics Seminar <i>KTH (virtual)</i>	Feb 2023
Combinatorial Algebra meets Algebraic Combinatorics (CAAC) conference <i>University of Waterloo</i>	Jan 2023
Special Session on Topological and Combinatorial Methods in Commutative Algebra <i>JMM Boston</i>	Jan 2023
Algebra and Geometry Seminar <i>Queen's University</i>	Nov 2022
Commutative Algebra Seminar <i>University of Michigan, Ann Arbor (virtual)</i>	Nov 2022
Commutative Algebra and Algebraic Geometry Seminar <i>University of Minnesota, Twin Cities</i>	Oct 2022

Expository Talks

A tour of Stanley–Reisner theory <i>AWM Undergraduate Seminar, Northeastern University</i>	Sept 2025
When are invariant rings . . . ? <i>Commutative Algebra Reading Seminar, University of Nebraska</i>	Nov 2023
Introduction to FI modules <i>Student Summer Representation Theory Seminar, University of Minnesota</i>	Aug 2023
Stabilization properties of chains of symmetric ideals <i>Student Combinatorics and Algebra seminar, University of Minnesota</i>	Mar 2022
Invariant theory and fixed quotients of polynomial rings <i>Oral/candidacy exam presentation</i>	Oct 2021
Gorenstein rings in the context of Stanley–Reisner theory  <i>Topics in Combinatorics course, University of Minnesota</i>	Apr 2021
Invariant rings, Hilbert series, and reflection groups  <i>Graduate Online Combinatorics Colloquium (GOCC)</i>	Nov 2020
The exchange property of Coxeter groups <i>Reading group on the combinatorics of Coxeter groups</i>	Jul 2020

Organizing

Northeastern University RTG REU	2026
Reviewed applications for admission	
Organized events and professional development workshops for REU participants	
AMS Special Session	2026
Spring 2026 Eastern Sectional Meeting at Boston College	

Session title: Group actions and combinatorics in algebraic geometry and commutative algebra

Seminars at Northeastern University

Geometry, Physics, and Representation Theory (GPRT) Seminar 2024-26

Geometry, Algebra, Singularities, and Combinatorics (GASC) Seminar 2024-26

Minnesota Research Workshop in Algebra and Combinatorics

A weeklong workshop for research and network building among University of Minnesota graduate students, alumni, and current/former postdocs in algebra and combinatorics. 2023-24

Directed Reading Program at University of Minnesota

2022-23

A semesterly program which pairs undergraduate students with graduate student mentors to explore a mathematical reading together.

Student Combinatorics and Algebra Seminar

2021-22

Outreach and Mentoring

Summer reading in commutative algebra

2026

With Northeastern PhD Student Erika Beserra

Bridge to Calculus at Northeastern University

2026

Contributed questions for mock AP exam for Calculus Field Day event

Mathematics Project at Minnesota

2020-23

An annual, weeklong workshop for underrepresented students interested in mathematics

Organized sessions on imposter syndrome, math puzzles, and future career opportunities

Guided small group of participants through reading and presenting on a mathematical article

One-on-one mentor for small group of participants

Directed Reading Program at University of Minnesota

2021, 23

Mentor for the following undergraduate reading projects:

Ideals, Varieties, and Algorithms by Cox, Little, O'Shea (Spring 2023)

The Finite Simple Groups by Wilson (Fall 2021)

Software

MatrixFactorizations package for Macaulay2

In-progress, current code available on Github 

Co-authored with David Favero, Timothy Tribone, and Keller VandeBogert

Supplemental functions for InvariantRing package in Macaulay2

Publicly available on Github 

Awards

Jean Roberts Fellowship (\$5000 award for research outcomes and service to the department) 2024

JMM Travel Grant 2023

AWM Travel Grant 2023

Nomination for SLMath Summer School 2022

University of Minnesota First Year Summer Fellowship 2019

University of Minnesota Professional Development Funding 2019

University of Minnesota Outstanding Graduate in Mathematics 2017

Industry Experience

PeopleNet Communications, Inc.

Technical Operations Engineer, Level 2

2017-19

Conducted long-term technical investigation for problems with on-board computers for commercial vehicles. Searched device logs, data pathways, and source code to diagnose issues with cell, Wi-Fi, messages, and engine data processing. Communicated to customers with varying levels of technical knowledge.

Skills

Software/programming

Macaulay2, SageMath, Mathematica, C++, Python, SQL

Languages

Fluent: English, Russian

Conversational: French